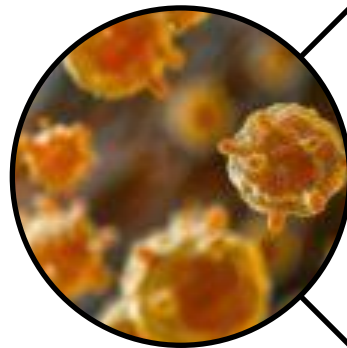


# **connecting people with science**



**Antimicrobial  
Resistance of Bacterial  
Pathogens – Africa**

7-13 March 2026

NICD, Johannesburg,  
South Africa

**Collins Kigen**

CP2 – Introduction to Command line

March 2026

# Navigation

| Command | Description                          |
|---------|--------------------------------------|
|         | Print Current Working Directory      |
|         | List Files and Directories           |
|         | Long list + human-readable file size |
|         | Show hidden files                    |
|         | Change directory                     |
|         | Move up one level                    |
|         | Go to home directory                 |

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# Navigation

| Command | Description                          |
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| pwd     | Print Current Working Directory      |
| ls      | List Files and Directories           |
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| Command | Description                          |
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| pwd     | Print Current Working Directory      |
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|         | Go to home directory                 |

# Navigation

| Command | Description                          |
|---------|--------------------------------------|
| pwd     | Print Current Working Directory      |
| ls      | List Files and Directories           |
| ls -lh  | Long list + human-readable file size |
|         | Show hidden files                    |
|         | Change directory                     |
|         | Move up one level                    |
|         | Go to home directory                 |

# Navigation

| Command | Description                          |
|---------|--------------------------------------|
| pwd     | Print Current Working Directory      |
| ls      | List Files and Directories           |
| ls -lh  | Long list + human-readable file size |
|         | Show hidden files                    |
|         | Change directory                     |
|         | Move up one level                    |
|         | Go to home directory                 |



# Navigation

| Command | Description                          |
|---------|--------------------------------------|
| pwd     | Print Current Working Directory      |
| ls      | List Files and Directories           |
| ls -lh  | Long list + human-readable file size |
| ls -a   | Show hidden files                    |
|         | Change directory                     |
|         | Move up one level                    |
|         | Go to home directory                 |

# Navigation

| Command | Description                          |
|---------|--------------------------------------|
| pwd     | Print Current Working Directory      |
| ls      | List Files and Directories           |
| ls -lh  | Long list + human-readable file size |
| ls -a   | Show hidden files                    |
|         | Change directory                     |
|         | Move up one level                    |
|         | Go to home directory                 |

# Navigation

| Command  | Description                          |
|----------|--------------------------------------|
| pwd      | Print Current Working Directory      |
| ls       | List Files and Directories           |
| ls -lh   | Long list + human-readable file size |
| ls -a    | Show hidden files                    |
| cd <dir> | Change directory                     |
|          | Move up one level                    |
|          | Go to home directory                 |

# Navigation

| Command  | Description                          |
|----------|--------------------------------------|
| pwd      | Print Current Working Directory      |
| ls       | List Files and Directories           |
| ls -lh   | Long list + human-readable file size |
| ls -a    | Show hidden files                    |
| cd <dir> | Change directory                     |
|          | Move up one level                    |
|          | Go to home directory                 |

# Navigation

| Command  | Description                          |
|----------|--------------------------------------|
| pwd      | Print Current Working Directory      |
| ls       | List Files and Directories           |
| ls -lh   | Long list + human-readable file size |
| ls -a    | Show hidden files                    |
| cd <dir> | Change directory                     |
| cd ..    | Move up one level                    |
|          | Go to home directory                 |

# Navigation

| Command  | Description                          |
|----------|--------------------------------------|
| pwd      | Print Current Working Directory      |
| ls       | List Files and Directories           |
| ls -lh   | Long list + human-readable file size |
| ls -a    | Show hidden files                    |
| cd <dir> | Change directory                     |
| cd ..    | Move up one level                    |
|          | Go to home directory                 |

# Navigation

| Command  | Description                          |
|----------|--------------------------------------|
| pwd      | Print Current Working Directory      |
| ls       | List Files and Directories           |
| ls -lh   | Long list + human-readable file size |
| ls -a    | Show hidden files                    |
| cd <dir> | Change directory                     |
| cd ..    | Move up one level                    |
| cd ~     | Go to home directory                 |

# Files and Directory Management

| Command | Description               |
|---------|---------------------------|
|         | Create a directory        |
|         | Create empty file         |
|         | Open editor               |
|         | Copy File                 |
|         | Move or rename file       |
|         | Remove File               |
|         | Remove Directory and File |
|         | Copy File                 |



# Files and Directory Management

| Command     | Description               |
|-------------|---------------------------|
| mkdir <dir> | Create a directory        |
|             | Create empty file         |
|             | Open editor               |
|             | Copy File                 |
|             | Move or rename file       |
|             | Remove File               |
|             | Remove Directory and File |
|             | Copy File                 |

# Files and Directory Management

| Command     | Description               |
|-------------|---------------------------|
| mkdir <dir> | Create a directory        |
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|             | Copy File                 |
|             | Move or rename file       |
|             | Remove File               |
|             | Remove Directory and File |
|             | Copy File                 |

# Files and Directory Management

| Command      | Description               |
|--------------|---------------------------|
| mkdir <dir>  | Create a directory        |
| touch <file> | Create empty file         |
|              | Open editor               |
|              | Copy File                 |
|              | Move or rename file       |
|              | Remove File               |
|              | Remove Directory and File |
|              | Copy File                 |

# Files and Directory Management

| Command      | Description               |
|--------------|---------------------------|
| mkdir <dir>  | Create a directory        |
| touch <file> | Create empty file         |
|              | Open editor               |
|              | Copy File                 |
|              | Move or rename file       |
|              | Remove File               |
|              | Remove Directory and File |
|              | Copy File                 |

# Files and Directory Management

| Command      | Description               |
|--------------|---------------------------|
| mkdir <dir>  | Create a directory        |
| touch <file> | Create empty file         |
| nano <file>  | Open editor               |
|              | Copy File                 |
|              | Move or rename file       |
|              | Remove File               |
|              | Remove Directory and File |
|              | Copy File                 |

# Files and Directory Management

| Command      | Description               |
|--------------|---------------------------|
| mkdir <dir>  | Create a directory        |
| touch <file> | Create empty file         |
| nano <file>  | Open editor               |
|              | Copy File                 |
|              | Move or rename file       |
|              | Remove File               |
|              | Remove Directory and File |
|              | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
|                 | Move or rename file       |
|                 | Remove File               |
|                 | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
|                 | Move or rename file       |
|                 | Remove File               |
|                 | Remove Directory and File |
|                 | Copy File                 |



# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
|                 | Remove File               |
|                 | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
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| cp <src> <dest> | Copy File                 |
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|                 | Remove File               |
|                 | Remove Directory and File |
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# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
| rm <file>       | Remove File               |
|                 | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
| rm <file>       | Remove File               |
|                 | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
| rm <file>       | Remove File               |
| rm -r <dir>     | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
| rm <file>       | Remove File               |
| rm -r <dir>     | Remove Directory and File |
|                 | Copy File                 |

# Files and Directory Management

| Command         | Description               |
|-----------------|---------------------------|
| mkdir <dir>     | Create a directory        |
| touch <file>    | Create empty file         |
| nano <file>     | Open editor               |
| cp <src> <dest> | Copy File                 |
| mv <src> <dest> | Move or rename file       |
| rm <file>       | Remove File               |
| rm -r <dir>     | Remove Directory and File |
| cp <src> <dest> | Copy File                 |

# Viewing Files

| Command | Description                               |
|---------|-------------------------------------------|
|         | First 10 lines                            |
|         | Last 10 line                              |
|         | Scroll large files one page at a time     |
|         | Similar to less but only forward move     |
|         | View entire file at once                  |
|         | View compressed file like sample.fastq.gz |



# Viewing Files

| Command           | Description                               |
|-------------------|-------------------------------------------|
| head -n 10 <file> | First 10 lines                            |
|                   | Last 10 line                              |
|                   | Scroll large files one page at a time     |
|                   | Similar to less but only forward move     |
|                   | View entire file at once                  |
|                   | View compressed file like sample.fastq.gz |

# Viewing Files

| Command           | Description                               |
|-------------------|-------------------------------------------|
| head -n 10 <file> | First 10 lines                            |
|                   | Last 10 line                              |
|                   | Scroll large files one page at a time     |
|                   | Similar to less but only forward move     |
|                   | View entire file at once                  |
|                   | View compressed file like sample.fastq.gz |

# Viewing Files

| Command           | Description                               |
|-------------------|-------------------------------------------|
| head -n 10 <file> | First 10 lines                            |
| tail -n 10 <file> | Last 10 line                              |
|                   | Scroll large files one page at a time     |
|                   | Similar to less but only forward move     |
|                   | View entire file at once                  |
|                   | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                               |
|--------------------------------------|-------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                            |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                              |
|                                      | Scroll large files one page at a time     |
|                                      | Similar to less but only forward move     |
|                                      | View entire file at once                  |
|                                      | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                               |
|--------------------------------------|-------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                            |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                              |
| <code>less &lt;file&gt;</code>       | Scroll large files one page at a time     |
|                                      | Similar to less but only forward move     |
|                                      | View entire file at once                  |
|                                      | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                               |
|--------------------------------------|-------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                            |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                              |
| <code>less &lt;file&gt;</code>       | Scroll large files one page at a time     |
|                                      | Similar to less but only forward move     |
|                                      | View entire file at once                  |
|                                      | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                               |
|--------------------------------------|-------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                            |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                              |
| <code>less &lt;file&gt;</code>       | Scroll large files one page at a time     |
| <code>more &lt;file&gt;</code>       | Similar to less but only forward move     |
|                                      | View entire file at once                  |
|                                      | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                               |
|--------------------------------------|-------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                            |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                              |
| <code>less &lt;file&gt;</code>       | Scroll large files one page at a time     |
| <code>more &lt;file&gt;</code>       | Similar to less but only forward move     |
|                                      | View entire file at once                  |
|                                      | View compressed file like sample.fastq.gz |



# Viewing Files

| Command           | Description                               |
|-------------------|-------------------------------------------|
| head -n 10 <file> | First 10 lines                            |
| tail -n 10 <file> | Last 10 line                              |
| less <file>       | Scroll large files one page at a time     |
| more <file>       | Similar to less but only forward move     |
| cat <file>        | View entire file at once                  |
|                   | View compressed file like sample.fastq.gz |

# Viewing Files

| Command                              | Description                                            |
|--------------------------------------|--------------------------------------------------------|
| <code>head -n 10 &lt;file&gt;</code> | First 10 lines                                         |
| <code>tail -n 10 &lt;file&gt;</code> | Last 10 line                                           |
| <code>less &lt;file&gt;</code>       | Scroll large files one page at a time                  |
| <code>more &lt;file&gt;</code>       | Similar to less but only forward move                  |
| <code>cat &lt;file&gt;</code>        | View entire file at once                               |
|                                      | View compressed file like <code>sample.fastq.gz</code> |

# Viewing Files

| Command           | Description                               |
|-------------------|-------------------------------------------|
| head -n 10 <file> | First 10 lines                            |
| tail -n 10 <file> | Last 10 line                              |
| less <file>       | Scroll large files one page at a time     |
| more <file>       | Similar to less but only forward move     |
| cat <file>        | View entire file at once                  |
| zcat <file.gz>    | View compressed file like sample.fastq.gz |

# Pipe

A pipe (|) is used to connect multiple commands. It takes the output from the previous command and “pipes” it into the input of the following command.

`wc -l example.txt` to count the number of lines in a file

`ls` to list the content of a directory

`ls | wc -l` to count the number of lines in a file

# Redirection

Another important character is the greater than sign `>`. This tells the command line to redirect the output to a file, rather than just printing it to the screen (STDOUT)

`ls` to count the number of lines in a file

`ls > content.txt` to count the number of lines in a file

# Append

Append means simply **add the data to the file without erasing existing data.**

`echo "Hello" > greeting.txt` to save first line to a file

`echo "How are you" >> greeting.txt` to add more data to existing file

# Wildcards

*Wildcards are special characters used to match patterns in filenames.*

| Wildcard | Description                                            | Example                     |
|----------|--------------------------------------------------------|-----------------------------|
| *        | Matches any number of characters                       | ls *.fastq                  |
| ?        | Matches exactly one character                          | ls sample?.txt              |
| [abc]    | Matches any one character inside the brackets          | ls file[abc].txt            |
| [a-z]    | Matches any character in the specified range           | ls file[a-z].txt            |
| [0-9]    | Matches any digit from 0 to 9                          | ls sample[0-9].fastq        |
| [!abc]   | Matches any character except those listed              | ls file[!abc].txt           |
| ^pattern | Matches lines starting with a pattern (used with grep) | grep "^ATG" sequences.fasta |
| {a,b,c}  | Matches any of the listed patterns (brace expansion)   | echo {01..12}               |

# Searching and counting

| Command | Description                     |
|---------|---------------------------------|
|         | Count lines                     |
|         | Count characters                |
|         | Search for text                 |
|         | Count the number of occurrences |
|         | Invert the match                |
|         | Sort and count unique           |



# Searching and counting

| Command | Description                     |
|---------|---------------------------------|
| wc -l   | Count lines                     |
|         | Count characters                |
|         | Search for text                 |
|         | Count the number of occurrences |
|         | Invert the match                |
|         | Sort and count unique           |

# Searching and counting

| Command | Description                     |
|---------|---------------------------------|
| wc -l   | Count lines                     |
|         | Count characters                |
|         | Search for text                 |
|         | Count the number of occurrences |
|         | Invert the match                |
|         | Sort and count unique           |

# Searching and counting

| Command | Description                     |
|---------|---------------------------------|
| wc -l   | Count lines                     |
| WC -C   | Count characters                |
|         | Search for text                 |
|         | Count the number of occurrences |
|         | Invert the match                |
|         | Sort and count unique           |

# Searching and counting

| Command | Description                     |
|---------|---------------------------------|
| wc -l   | Count lines                     |
| wc -c   | Count characters                |
|         | Search for text                 |
|         | Count the number of occurrences |
|         | Invert the match                |
|         | Sort and count unique           |

# Searching and counting

| Command               | Description                     |
|-----------------------|---------------------------------|
| wc -l                 | Count lines                     |
| wc -c                 | Count characters                |
| grep "pattern" <file> | Search for text                 |
|                       | Count the number of occurrences |
|                       | Invert the match                |
|                       | Sort and count unique           |

# Searching and counting

| Command               | Description                     |
|-----------------------|---------------------------------|
| wc -l                 | Count lines                     |
| wc -c                 | Count characters                |
| grep "pattern" <file> | Search for text                 |
|                       | Count the number of occurrences |
|                       | Invert the match                |
|                       | Sort and count unique           |

# Searching and counting

| Command                                     | Description                     |
|---------------------------------------------|---------------------------------|
| <code>wc -l</code>                          | Count lines                     |
| <code>wc -c</code>                          | Count characters                |
| <code>grep "pattern" &lt;file&gt;</code>    | Search for text                 |
| <code>grep -c "pattern" &lt;file&gt;</code> | Count the number of occurrences |
|                                             | Invert the match                |
|                                             | Sort and count unique           |

# Searching and counting

| Command                  | Description                     |
|--------------------------|---------------------------------|
| wc -l                    | Count lines                     |
| wc -c                    | Count characters                |
| grep "pattern" <file>    | Search for text                 |
| grep -c "pattern" <file> | Count the number of occurrences |
|                          | Invert the match                |
|                          | Sort and count unique           |



# Searching and counting

| Command                  | Description                     |
|--------------------------|---------------------------------|
| wc -l                    | Count lines                     |
| wc -c                    | Count characters                |
| grep "pattern" <file>    | Search for text                 |
| grep -c "pattern" <file> | Count the number of occurrences |
| grep -v "pattern" <file> | Invert the match                |
|                          | Sort and count unique           |

# Searching and counting

| Command                                     | Description                     |
|---------------------------------------------|---------------------------------|
| <code>wc -l</code>                          | Count lines                     |
| <code>wc -c</code>                          | Count characters                |
| <code>grep "pattern" &lt;file&gt;</code>    | Search for text                 |
| <code>grep -c "pattern" &lt;file&gt;</code> | Count the number of occurrences |
| <code>grep -v "pattern" &lt;file&gt;</code> | Invert the match                |
|                                             | Sort and count unique           |

# Searching and counting

| Command                                     | Description                     |
|---------------------------------------------|---------------------------------|
| <code>wc -l</code>                          | Count lines                     |
| <code>wc -c</code>                          | Count characters                |
| <code>grep "pattern" &lt;file&gt;</code>    | Search for text                 |
| <code>grep -c "pattern" &lt;file&gt;</code> | Count the number of occurrences |
| <code>grep -v "pattern" &lt;file&gt;</code> | Invert the match                |
| <code>sort file.txt   uniq -c</code>        | Sort and count unique           |

# Productivity Shortcuts

| Command | Description                         |
|---------|-------------------------------------|
|         | Autocomplete                        |
|         | Stop running process                |
|         | Clear screen                        |
|         | Previous command                    |
|         | Search command history              |
|         | Run last command                    |
|         | Go to the start of the command line |
|         | Go to the end of the command line   |

# Productivity Shortcuts

| Command | Description                         |
|---------|-------------------------------------|
| Tab     | Autocomplete                        |
|         | Stop running process                |
|         | Clear screen                        |
|         | Previous command                    |
|         | Search command history              |
|         | Run last command                    |
|         | Go to the start of the command line |
|         | Go to the end of the command line   |

# Productivity Shortcuts

| Command | Description                         |
|---------|-------------------------------------|
| Tab     | Autocomplete                        |
|         | Stop running process                |
|         | Clear screen                        |
|         | Previous command                    |
|         | Search command history              |
|         | Run last command                    |
|         | Go to the start of the command line |
|         | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
|          | Clear screen                        |
|          | Previous command                    |
|          | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
|          | Clear screen                        |
|          | Previous command                    |
|          | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |



# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
|          | Previous command                    |
|          | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
|          | Previous command                    |
|          | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command                                                                           | Description                         |
|-----------------------------------------------------------------------------------|-------------------------------------|
| Tab                                                                               | Autocomplete                        |
| Ctrl + C                                                                          | Stop running process                |
| Ctrl + L                                                                          | Clear screen                        |
|  | Previous command                    |
|                                                                                   | Search command history              |
|                                                                                   | Run last command                    |
|                                                                                   | Go to the start of the command line |
|                                                                                   | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
|          | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
|          | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
| !!       | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
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| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
| !!       | Run last command                    |
|          | Go to the start of the command line |
|          | Go to the end of the command line   |



# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
| !!       | Run last command                    |
| Ctrl + a | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
| !!       | Run last command                    |
| Ctrl + a | Go to the start of the command line |
|          | Go to the end of the command line   |

# Productivity Shortcuts

| Command  | Description                         |
|----------|-------------------------------------|
| Tab      | Autocomplete                        |
| Ctrl + C | Stop running process                |
| Ctrl + L | Clear screen                        |
| ↑        | Previous command                    |
| Ctrl + R | Search command history              |
| !!       | Run last command                    |
| Ctrl + a | Go to the start of the command line |
| Ctrl + e | Go to the end of the command line   |

# Help

| Command          | Description                                |
|------------------|--------------------------------------------|
| man <command>    | Full manual for any command                |
| <command> --help | Quick Help                                 |
| history          | Show previous commands that have been run  |
| Info <command>   | Show detailed information about a command. |
|                  |                                            |
|                  |                                            |
|                  |                                            |

# Acknowledgements and Partners



# thank you!

Please contact [wellcomeconnectingscience.org](https://wellcomeconnectingscience.org)  
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